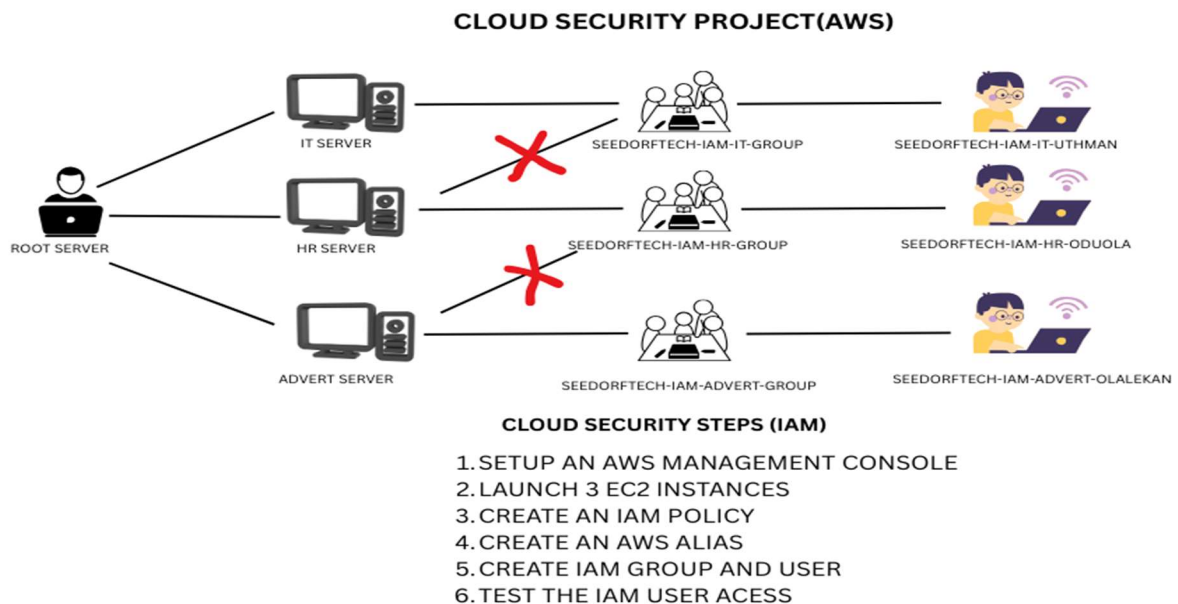


AWS IAM Cloud Security Project

1. Project Overview

I completed this project on cloud security controls in Amazon Web Services (AWS), focusing on Identity and Access Management (IAM). The goal was to create a least-privilege policy, attach it to a user group, and verify that the policy correctly restricts actions on three Amazon EC2 instances (IT, Human Resource Management and Advert).



2. Tools & Concepts

- AWS IAM – users, groups, policies, account alias
- Amazon EC2 – instance tagging and lifecycle actions
- JSON policy syntax – Effect, Action, Resource
- Principle of least privilege and policy testing

3. Tagging Strategy

I applied a descriptive tag to each EC2 instance:

Instance | Tag Key | Tag Value

HRM | Environment | Human Resource Management

IT | Environment | Information Technology

Advert | Environment | Advert

Instances (3) Info										
Find Instance by attribute or tag (case-sensitive)										
☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
☐	MyEC2TServer	i-0bd8f281d4447bec1	Running	t3.micro	3/3 checks passed	View alarms +	us-east-2c	ec2-18-191-57-208.us-...	18.191.57.208	-
☐	MyEC2HServer	i-02cda658b6150d130	Running	t3.micro	3/3 checks passed	View alarms +	us-east-2c	ec2-18-188-249-201.us...	18.188.249.201	-
☐	MyEC2ADVERTServer	i-0abb7dd49315c57b2	Running	t3.micro	3/3 checks passed	View alarms +	us-east-2c	ec2-18-118-99-195.us-...	18.118.99.195	-

4. Creating the IAM Policy

I authored the following JSON policy to block instance stop/start actions on the audit server but allow those actions on the sales server:

Permissions defined in this policy [Info](#)

[Copy](#)
[Edit](#)
[Summary](#)
[JSON](#)

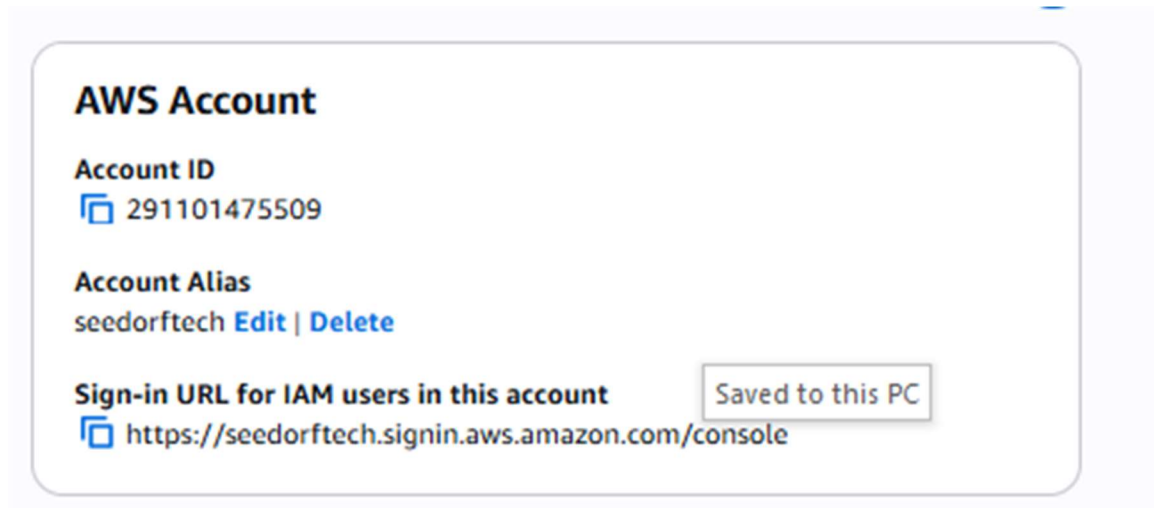
Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "ec2:*",
7       "Resource": "*",
8       "Condition": {
9         "StringEquals": {
10          "ec2:ResourceTag/Env": "Audit"
11        }
12      }
13    },
14    {
15      "Effect": "Allow",
16      "Action": "ec2:Describe*",
17      "Resource": "*"
18    },
19    {
20      "Effect": "Deny",
21      "Action": [
22        "ec2:DeleteTags",
23        "ec2:CreateTags"
24      ],
25      "Resource": "*"
26    }
27  ]
28 }
```

5. Account Alias

I set a memorable account alias to replace the default numeric URL, making sign-in easier for team members.



6. IAM Users & Groups

1. Created an IAM user group called Developers.
2. Attached the **CybertechAuditEnvPolicy** policy to the group.
3. Added individual IAM users who require controlled EC2 access.

Users (4) [Info](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Q Search

☐	User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key ID	At
☐	Seedorftech-IAM-Advert-Olalekan	/	1	-	-	6 days	-	-	-
☐	Seedorftech-IAM-HumanResourceManagement-Oduola	/	1	-	-	6 days	-	-	-
☐	Seedorftech-IAM-InformationTechnology-Uthman	/	1	-	-	6 days	-	-	-
☐	Seedorftech-IAM-Saheed	/	0	3 hours ago	-	6 days	3 hours ago	-	-

7. Logging in as an IAM User

IAM users can sign in through:

- AWS Management Console (using the new alias URL)
- AWS CLI via programmatic keys

AWS Search [Alt+S] Europe (Stockholm) cybertech-Audit-Ali @ cybertechusers

Console Home [Info](#) [Reset to default layout](#) [Add widgets](#)

Recently visited [Info](#)

No recently visited services

Explore one of these commonly visited AWS services.

[EC2](#) [S3](#) [Aurora and RDS](#) [Lambda](#)

[View all services](#)

Applications (0) [Info](#) [Create application](#)

Region: Europe (Stockholm)

Select Region: eu-north-1 (Current Region) [Find applications](#)

< 1 >

Name	Description	Region	Originati
Access denied to servicecatalog:ListApplications			
Diagnose with Amazon Q			

[Go to myApplications](#)

Welcome to AWS [Getting started with](#)

AWS Health [Info](#)

Cost and usage [Info](#)

Current month costs [Cost breakdown](#)

8. Testing the Policy

Test	Action		Expected	Result		Actual	Result
Stop	audit	instance	Denied	Access	denied	error	displayed
Stop	sales	instance	Allowed	Instance	stopped		successfully
Start	audit	instance	Denied	Access	denied	error	displayed
Start sales instance Allowed Instance started successfully							

•

IAM Dashboard Info

Security recommendations0

⛔ Access denied

You don't have permission to `iam:GetAccountSummary`. To request access, copy the following text and send it to your AWS administrator. [Learn more about troubleshooting access denied errors.](#)

User: arn:aws:iam::288645738432:user/cybertech-Audit-Ali

Action: iam:GetAccountSummary

Context: no identity-based policy allows the action

Diagnose with Amazon Q

⛔ Access denied

You don't have permission to `iam:ListMFADevices`. To request access, copy the following text and send it to your AWS administrator. [Learn more about troubleshooting access denied errors.](#)

User: arn:aws:iam::288645738432:user/cybertech-Audit-Ali

Action: iam:ListMFADevices

Context: no identity-based policy allows the action

⛔ Failed to stop the instance i-0b9626f2835a5f977

You are not authorized to perform this operation. User: arn:aws:iam::288645738432:user/cybertech-Audit-Ali is not authorized to perform: ec2:StopInstances on resource: arn:aws:ec2:us-east-1:288645738432:instance/i-0b9626f2835a5f977 because no identity-based policy allows the ec2:StopInstances action. Encoded authorization failure message: sAveffO1FXITMRJ52xofhO4bpSGLZ7IFqT-w1DqTxLoat_IV4T5r87OVMHxzfSSmG4VXfVa2cDWhfQ1uQOshFk8kUgfZhoC9Tn_lhCnKyVswG-blgkFFw9OLb4-2CT9TP841I8ni9O6EaSTAt0ztgtsh2t1VXnMzhUuPZeqMg3Ajek6wr21590c0le6YNIJRMmocGw8Gd_xH9dlfdp-4ilwF0H1uwTpAt1QHgqr8FbW3kJOue7UMNuHGzJzgCiiA30Q5pqHwW1_ALxzQAulhMpW9I6xdEiwRRLOFQG_W-dDTm6Z7YNPth09woBuuiXzLbc13Mz5-ciFyB_0PBgVz8xKMy4kuncL_zDmTsNfyvOsszpKwvYxm8Du71JN3rSTmBylKAIW0cA0nzcOFVWIQ9kigwtbT-gocongZGILqW2RxlqW8XBC420EN3ILZ-URE-xyZCS1palfmclLRiDvCBZJNWPPr4rK_OWgaADMkzFgQjk7xmnnNmFi2mgHr5Aw46VvHCp-pqURgYZ_Oa8xBnq7y6lulZ_MxlUub20pZDhB0SynHoNGRm0MRzRy1etyCipN14aBfVWB12hSkMoXobUzawc9Rat-Pk_vPSbim-Opv1riuGTCT-YJtIZ6FSYUFEcxnmX6B3IFHqgGT0p6sDazlXMCX3g2yqEqbTNSf-CIWsZbvA7Ck5m5yf8nt1po_-8fBDM67ZwyEe21nGxikjEmj_nMMMMfQVPqk46tBHdwrYHLxQQA2b-cG58fOMUsv_eHnT1xBHLtkVMB7VxU-_Scg_KyhwQIMrGy4NR4Ho_RumoDyK8aWNLTixh-j1yTKrJk8ctDTf11IQzyfSAKRTQp_-j1QBrzXEA47t8IXxAwqRD3vWCNE3PL3YfwcYpRHP

Diagnose with Amazon Q

Instances (1/2) Info

Find Instance by attribute or tag (case-sensitive)

All states

1

i-0b9626f2835a5f977 (Cybertech-sales-Ali)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags